

DHEERAJ VERMA

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[LinkedIn](#) | [GitHub](#) | [Portfolio](#)

PROFESSIONAL SUMMARY

AI/ML Engineer and B.Tech Computer Science graduate (AI/ML specialization) with hands-on experience in Python, SQL, data preprocessing, feature engineering and model evaluation. Built end-to-end ML and Generative AI projects covering NLP, RAG, multi-agent systems and MLOps (MLflow, DVC, Docker, AWS), and currently works as a Trainee Decision Scientist on structured business datasets.

TECHNICAL SKILLS

Programming: Python, SQL, C++

Machine Learning: Scikit-learn, LightGBM, Regression, Classification, Clustering, Feature Engineering, Model Evaluation, Imbalanced Learning

Data & Analysis: Pandas, NumPy, Matplotlib, Data Preprocessing, Data Profiling, Statistical Analysis

NLP & Generative AI: Hugging Face Transformers, TF-IDF, Embeddings, LLMs, RAG, AI Agents, LangChain, Langflow

MLOps, APIs & Databases: MLflow, DVC, Docker, AWS, Git, GitHub, CI/CD, Flask, FastAPI, REST APIs, PostgreSQL, MongoDB

EXPERIENCE

Trainee Decision Scientist, Mu Sigma

Apr 2026 – Present

- Used Python, SQL and Pandas to clean, transform, validate and analyze structured datasets, running data quality checks for missing values, inconsistencies, outliers and integrity issues.
- Developed analytical workflows covering data profiling, feature preparation, statistical analysis and business metric evaluation.
- Applied clustering, customer segmentation, outlier detection and KPI analysis to derive business insights.
- Worked in Agile project environments, translating business requirements into analytical tasks and communicating findings to stakeholders.

Data Science and Analytics Intern, Proxenix (Remote)

Apr 2025 – Jun 2025

- Cleaned, transformed and validated structured datasets using Python, SQL and Pandas, and assessed data quality.
- Performed data profiling, feature preparation and statistical analysis to support analytical problem solving.

PROJECTS

YouTube Sentiment Analysis & MLOps Pipeline | Python, LightGBM, MLflow, DVC, Docker, AWS

Feb 2026 – May 2026

- Built an end-to-end NLP sentiment classification pipeline using TF-IDF features and LightGBM, including text preprocessing, feature engineering and train/test evaluation.
- Addressed class imbalance with SMOTE and evaluated the classifier with classification metrics, reaching approximately 80% accuracy.
- Tracked experiments and managed the model lifecycle with MLflow; versioned data and experiments with DVC; used Docker for reproducible environments.
- Designed the pipeline for AWS-based deployment, covering data preparation, training and model serving.

GenAI Fitness Assistant – Multi-Agent ML Project | Python, LLMs, Langflow, Streamlit, Astra DB

Nov 2025 – Jan 2026

- Built a multi-agent Generative AI application with Retrieval-Augmented Generation (RAG) that grounds LLM responses in information retrieved from a vector database (Astra DB).
- Designed modular agents with task-specific routing and separated retrieval, reasoning and user-facing components.

Image Colorization using Deep Learning | Python, U-Net, Conditional GAN

Apr 2026 – May 2026

- Built a deep learning image colorization model in Python using U-Net and a Conditional GAN.

Grid-World Q-Learning Agent | Python, Reinforcement Learning

Apr 2026 – May 2026

- Implemented a Q-learning agent from scratch, designing state-action representations, reward mechanisms and exploration-exploitation strategies, and analyzed convergence across training episodes.

EDUCATION

B.Tech, Computer Science & Engineering (AI/ML Specialization)

2026 | CGPA 7.1/10

Jaypee University of Engineering and Technology, Guna, Madhya Pradesh

CERTIFICATIONS & LEADERSHIP

- Machine Learning Scientist in Python (85 hours), DataCamp
- Oracle Cloud Infrastructure 2025 Certified Generative AI Professional; OCI 2025 AI Foundations Associate
- Google Developer Student Club Coordinator (Sep 2023 – Aug 2024): mentored 50+ students through technical workshops